

Research Article

A Graph Representation Learning-Based Method for Event Prediction

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Received 2 November 2024; Accepted 6 May 2025

Academic Editor: Peican Zhu

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With the continuous advancement of big data and artificial intelligence technologies, event prediction is increasingly being utilized across a multitude of domains. Predicting events allows for the exploration of the developmental trajectories and summarization of patterns associated with these events. However, events typically encompass a myriad of elements and intricate relationships, necessitating an enhancement in the precision of event prediction. However, the existing methods suffer from poor data quality, insufficient feature information, limited generalization capability of the models, and difficulties in evaluating prediction errors. This paper proposes a novel event prediction method based on graph representation learning, aiming to improve the accuracy of event prediction while reducing the time cost. By constructing causal graphs and introducing the script event simulation method, the architecture combines graph neural networks (GNNs) with BERT to simplify the event prediction process. Additionally, by combining GNNs with pretrained language models, a dynamic graph representation learning method is proposed. This means that a unified graph representation learning model can be built by following specific rules, thus predicting the development trajectory of events more accurately. The study evaluates the effectiveness of dynamic graph representation learning technology in a specific scenario, specifically in the context of employee career choices. By converting the career graph of employees into low-dimensional representations, the effectiveness of the dynamic graph representation learning method in predicting employee career decisions is validated. This innovation not only improves the accuracy of event prediction but also helps better understand and respond to complex event relationships in practical applications, providing decision-makers with more powerful information support. Therefore, this research has important theoretical and practical significance, providing valuable references for future studies in related fields.

Keywords: career choice prediction; event prediction; graph representation learning; heterogeneous graph

1. Introduction

In recent years, with the rapid advancement of big data and artificial intelligence, an increasing number of companies, research institutions, and even governments are eager to leverage these technologies to forecast the impact of current events on future developments and predict upcoming occurrences. Examples include predicting the popularity of trending topics on Weibo using text analysis, inferring subsequent events from

current contexts, and forecasting online public opinion using causal graphs [1]. Event prediction holds significant social and economic implications, capturing growing interest among researchers. Understanding event development patterns and simulating their evolution in real-world scenarios are pressing questions for many scholars [2].

Event prediction utilizes a variety of methods and models to foresee future events or trends. By anticipating potential events, decision-makers can formulate strategies to mitigate

uncertainties. For instance, predicting employee career paths can assist managers in effectively reducing turnover, enhancing a company's economic performance, and consequently improving social benefits [3]. Event prediction, through analyzing the development trajectories of events and summarizing their inherent patterns, offers substantial social and economic value.

However, event prediction is a complex task. Events often involve multiple factors, whose developments and changes are fraught with uncertainty and influenced by randomness and external factors. Moreover, complex interrelationships exist between events, necessitating a thorough analysis of related events' development trends and their impact on the focal event for accurate predictions. The accuracy of event predictions is also constrained by the quality of models and data; imperfections in either can lead to significant deviations.

In the current event prediction solutions, many data have missing, incorrect, or noisy issues, and it is impossible to obtain all relevant feature information, resulting in insufficient accuracy of the prediction model. At the same time, some prediction models have weak generalization ability on unknown data, making it difficult to explain the reasons for their prediction results, limiting their credibility and acceptance in practical applications.

Events' relationships can be depicted through a dense directed graph, and by clustering and generalizing events, one can form a graph describing event evolution patterns, known as a causal graph. Many learning tasks now require graph data, replete with rich relational information among elements. By mining knowledge from the intricate relationships between events, we can unveil their developmental patterns. A causal graph describes these relationships, allowing us to simulate subsequent changes in events through specific analytical methods.

By integrating big data and artificial intelligence technology, we can enhance the precision of predicting future event trends and impacts. In comparison to current research, our approach is adept at managing multiple variables, uncertainties, and intricate relationships, thereby enhancing the precision and dependability of event prognostication. The existing challenges that require attention revolve around the intricacy and unpredictability of forecasting events. Conventional techniques often encounter difficulties in accurately foretelling the future due to the involvement of randomness and external factors in events. Our methodology, involving the construction of causal graphs, event clustering, and summarization, along with the exploration of complex event interconnections, enables a better comprehension of event evolution patterns and facilitates future trend prediction. Moreover, through enhancing data and model quality, we can amplify the accuracy and consistency of event prediction. The depiction of event relationships using causal graphs enables a more effective simulation of subsequent event alterations, furnishing decision-makers with robust references and guidance. This will not only diminish uncertainty and improve predictive precision but also aid businesses and institutions in devising more efficacious strategies and decisions [4].

This paper introduces an event prediction method leveraging graph representation learning. By applying graph

representation techniques, it characterizes and induces developmental patterns and rules of events to conduct robust event predictions. The main contributions of this study are as follows:

1. Construction of a causal graph for the event dataset based on specific rules, integrating pretraining models to regularize events. We propose a methodological architecture combining graph neural networks (GNNs) with BERT to simulate script events.
2. Introduction of a dynamic graph representation learning method to address the complexity of real event nodes. Using specific patterns, complex nodes are sampled to construct a homogeneous graph model for precise event prediction.
3. Development of a dynamic bipartite graph representation learning approach applied to a case of worker career decisions. Its effectiveness is verified by mapping job situation graphs to low-dimensional spaces.

This study is organized as follows: Section 2 provides a comprehensive review of event prediction research and causal graph ontologies globally. Section 3 presents the script event prediction method using GNNs. Section 4 discusses the dynamic graph representation learning model, validated through worker career choice examples. Section 5 concludes the work presented in this paper.

2. Related Work

Research on event prediction employs diverse methodologies to forecast future outcomes. Common approaches include statistical analyses, machine learning, and causal graph ontologies.

1. Statistical analysis: Utilizes historical data and statistical models to predict future event probabilities and trends, employing techniques such as time series analysis, regression, and classification models. For instance, Javid Iqbal and Khalid Riaz analyzed the relationship between managerial tone and the future financial performance of banks using linear regression and time series models. They used current financial indicators as dependent variables and managerial tone as independent variables [5, 6]. Xie et al. [7] combined deep learning prediction with statistical analysis for quality control of time series wave data, enhancing the detection of anomalies in ocean data.
2. Machine learning: Employs algorithms to discover patterns from large datasets, predicting future events based on learned insights. Techniques include decision trees, neural networks, and support vector machines. For example, Castillo-Bot et al. [8, 9] applied machine learning for fog event prediction, demonstrating its utility in environmental applications. Similarly, "predicting stock market trends using machine learning techniques" analyzed stock market trends, offering insights into utilizing machine learning for financial predictions [10].
3. Causal graph ontologies: Aim to understand network dynamics' impact on predictive performance. Goyal et al.

[11] proposed a dynamic graph vector mapping model that predicts events through trend analysis of dynamic graphs. Peng et al. [12] introduced unsupervised extraction of event graph structures for mutual information analysis, enhancing homogeneous graph transformation. Sankar et al. [13] utilized dynamic neural networks combined with self-attention mechanisms for improved graph structure representation. Other notable contributions include Goodall R's element-weighted graph representation, which improved link prediction accuracy [14], and Li P's distance coding method enhancing event node representation efficiency [15]. Zheng et al. [16] proposed a supervised graph optimization representation, refining event node information for superior prediction accuracy. Le-Khac et al. [17] reviewed graph comparison learning, highlighting its future potential. Kazemi et al. [18] detailed the latest trends in dynamic graph representation, focusing on application and dataset analysis.

3. Script Event Prediction Method Based on GNN

This section proposes an event prediction method based on graph representation learning, by adopting graph representation learning techniques to characterize the developmental patterns of events and conduct event predictions, thus enhancing the accuracy of event prediction. In the following, we delineate the script event prediction, integrating GNNs and BERT for method construction and simulation inference.

3.1. Script Event Prediction Description. Script event prediction entails precise simulations of designated event nodes within specified contexts. This study combines pretraining models with GNNs to establish a predictive framework for script events. Given the necessity of linking node relationships through graph knowledge, language understanding via natural language processing and pretraining models is imperative. While skip-gram models require extensive training, they produce high-quality word vectors [19]. The BERT model excels in extracting representations from vast corpora, proving invaluable for complex scenarios [20]. GNNs facilitate node link modeling, crucial for data transmission and updates. Here, specialized variables are employed to connect event nodes, enhancing event map coverage and speeding up script event prediction [21]. This method's feasibility and efficacy are validated using industry-standard datasets, as illustrated in Figure 1.

It can be seen from Figure 1 that script event prediction can be described as follows: event key role A and event node chain $E_a = \{e_1, \dots, e_i, \dots, e_n\}$ around key role A, a single node $e_i = \{P_i(S_i, O_i, I_i)\}$, n is the total number of nodes, and I_i, O_i, S_i, P_i are indirect object, object, subject, and predicate. For the chain before and after a specific event node, it is necessary to match the best possible node from the development situation $C_a = \{c_1, \dots, c_m\}$ after the node, and the description of each possible node $c_j = \{P_j(S_j, O_j, I_j)\}$ is consistent with the context event node.

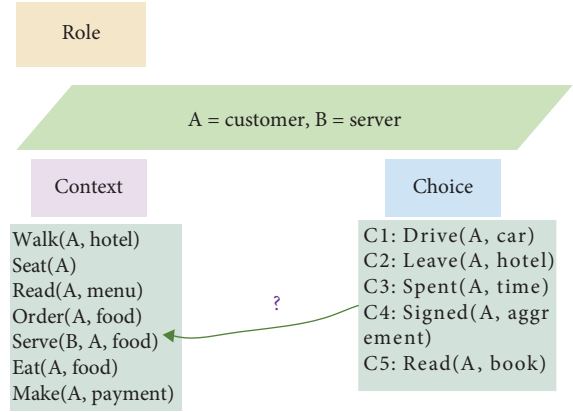


FIGURE 1: Script prediction case.

The event graph can be described by $G = \{V, E\}$, where $V = \{v_1, \dots, v_i, \dots, v_p\}$ is the set of nodes in the event graph, which are associated with a single event, and $E = \{l_1, \dots, l_i, \dots, l_q\}$ represents the edge set of the event graph, where the number of nodes is represented by p and q . Combining events in the training set to construct the event graph, the main logic for judging the event link relationship is whether the events occur sequentially. In order to improve the performance of the event composition, select the main character grammar and verbs of the event to be properly reserved, and construct the directed edge of the event graph as $v_i \rightarrow v_j$. The weight parameter ω is determined by Formula (1):

$$\omega_{ij} = \frac{\text{count}(v_i, v_j)}{\sum_k \text{count}(v_i, v_k)}, \quad (1)$$

where $\text{count}(v_i, v_j)$ represents the number of occurrences of the key element in the event node.

The final scripted event graph consists of over 6 million directed weighted edges, incorporating ~110,000 key elements. This structure enables the successful low-dimensional mapping of interaction information and link relationships among event nodes.

3.2. Scripted Event Simulation Method Integrating GNN and BERT. To simultaneously capture the structural and semantic aspects of the scripted event graph, it is necessary to design a method that integrates GNNs with the BERT model. The BERT model performs regularization of event nodes, while the GNN manages the linkage of node relationships. To enhance prediction speed, special variable links are introduced to the event nodes. This method encompasses components such as regularization, aggregation, reasoning, updating, mixing, and prediction. Regularization leverages contextual information for event vector processing [22, 23], whereas aggregation consolidates individual node attributes into a comprehensive event. Reasoning enhances the linkage of event relationships, updating involves refreshing event nodes in conjunction with link relationships, and mixing primarily addresses context representation alongside updated event aggregation processes. Prediction entails simulating future

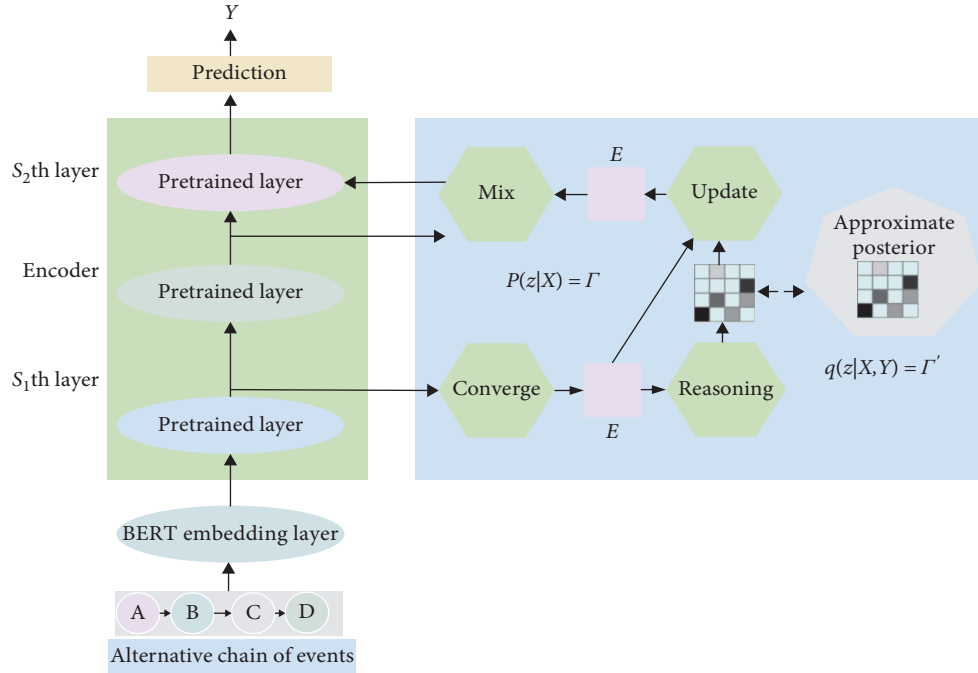


FIGURE 2: Architecture of script event simulation method combining graph neural network and BERT.

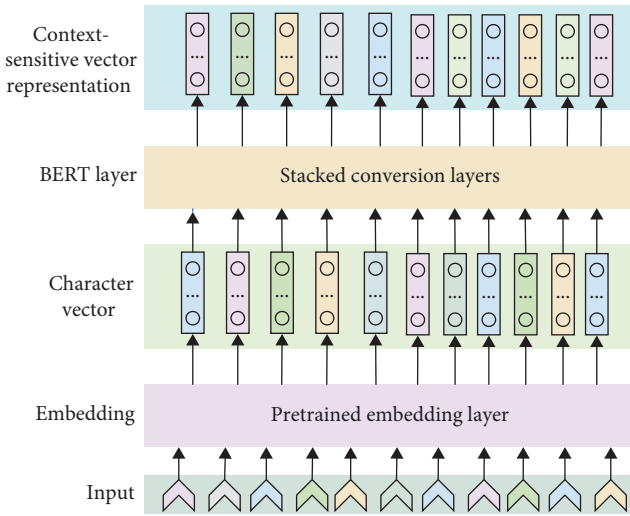


FIGURE 3: Pretraining model architecture.

events based on context analysis and validating whether input alternative event nodes are accurate or not [24, 25]. The overall architecture of this model is depicted in Figure 2.

3.2.1. Regularization. Regularization primarily employs BERT for event processing. Once the input information is initialized, it proceeds to the stack conversion layer to achieve the representation of character vectors. The model's specific framework is illustrated in Figure 3.

The initial vector derived from processing the input information comprises three components: segment, position, and character vectors. These character vectors are then fed into the stacked conversion layer, which consists of a feedforward network coupled with a residual connection. The multihead self-

attention layer is also part of this architecture, as detailed in Figure 4.

As depicted in the figure, the attention mechanism is formed by the value, key, and query vectors. To address the limitations inherent in conventional attention mechanisms, we adopt an enhanced attention mechanism as specified by Formula (2). This approach provides improved handling of complex relationships and dynamic variations within events, thereby effectively enhancing the prediction accuracy and robustness:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V, \quad (2)$$

where d_k corresponds to the vector dimension in key K . BERT combines the attention mechanism to obtain various semantic space information. The method flow is as follows: the first step is to divide the query Q , K , and V into h heads according to their dimension features $\frac{d_k}{h}$ and then divide a single head into a low-dimensional vector space, thus getting the semantic space data. See Formulas (3) and (4) for the processing of h heads:

$$\text{head}_i = \text{Attention}(QW_i^Q, KW_i^K, VW_i^V), \quad (3)$$

$$\text{Multihead}(Q, K, V) = \text{Concat}(\text{head}_1, \dots, \text{head}_h)W^0. \quad (4)$$

In this formula, d_{model} is the size of the hidden layer of the model, and $W^0 \in \mathbb{R}^{d_{\text{model}} \times h d_v}$ and $W^0 W_i^Q, W_i^K, W_i^V \in \mathbb{R}^{d_{\text{model}} \times d_k}$ refer to training-related parameters.

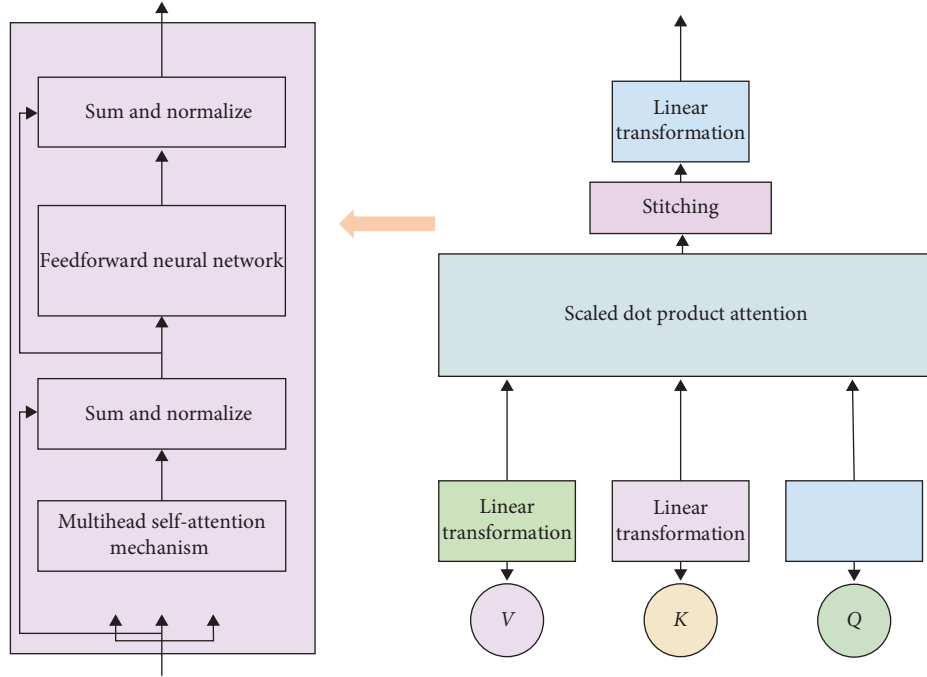


FIGURE 4: Single-layer structure of stacked conversion layers.

In order to prevent the gradient vanishing problem, a special residual processing is adopted, and the method is shown in the Formula (5):

$$\text{Residual}(Q, K, V) = \text{Layer norm}(V + \text{Multihead}(Q, K, V)). \quad (5)$$

The subsequent processing is completed by the feedforward network, and its specific processing method is shown in Formula (6):

$$\text{Transformer}(Q, K, V) = \text{GeLU}(\text{Residual}(Q, K, V)W_1 + b_1)W_2 + b_2. \quad (6)$$

GeLU is the Gaussian error linear unit. The output of pre-training is a vector of contextual information, which contains n transformation pair stacks.

3.2.2. Convergence. Due to the differences in the levels of information obtained by pretraining, this time S_1 layer is selected as the output and aggregated. Pooling converts the output of the pretrained S_1 layers into a vector representation of each event [26].

Given an event $X_i = \{x_1, \dots, x_{l_i}\}$ containing l_i characters, it is input to the stacked conversion layer of the previous S_1 layer to obtain a character vector, and the attention mechanism is used for event aggregation, and $q_i = \frac{1}{l_i} \sum h_{l_i}$ is described as the vector average of each character. The calculation method of the event i is shown in Formula (7):

$$e_i = \text{Multihead}(q_i, h_i^{(S_1)}, h_i^{(S_1)}). \quad (7)$$

Based on the above methods, all event nodes can be replaced by vectors, and event vectors can be described by matrices. $E' = \{e_1 \oplus \dots \oplus e_n\} \in \mathbb{R}^{n \times d}$, d in the matrix represents the vector dimension, n represents the number of events, and $\oplus$ represents matrix splicing. The processing formula for obtaining refined events combined with the attention mechanism is shown in Equation (8):

$$E = \text{Multihead}(E', E', E'). \quad (8)$$

3.2.3. Reasoning. Aiming at the problem that the event prediction takes too long and the coverage rate of the event map is low, special variables are added to process the event matrix. The main method is to use the fitting matrix to copy the matrix during training and directly use the fitting matrix for prediction when predicting, reducing Matrix search time [27].

The first step is to obtain the input event link matrix $q(z|X, Y) = \Gamma'$, where $\Gamma' \in \mathbb{R}^{n \times n}$, $\Gamma' = \omega_{ij}$, and then reason the set E represented by the event to obtain the link matrix between events. The process adds a special variable z to complete the matrix fitting. The detailed method is as shown in Formula (9):

$$p(z|X) = \Gamma = \text{softmax}\left(\frac{EW_1(EW_2)^T}{\sqrt{d}}\right), \quad (9)$$

where d represents the vector dimension and W_1, W_2 is the trainable vector.

3.2.4. Update. Events are updated based on the event link matrix. The event update mechanism used in different stages is different. The event update method is as shown in Formula (10):

$$E^{(U)'} = \begin{cases} \Gamma' E, & \text{training stage} \\ \Gamma E, & \text{testing stage} \end{cases}. \quad (10)$$

As shown in Formula (10), the event update method $E^{(U)'}$ is adjusted by the parameter Γ' during the training stage, while the fixed parameter Γ is adopted during the testing stage.

After obtaining the updated event, use Formula (11) combined with the attention mechanism to obtain the fine event representation, and the specific method is as follows:

$$E^{(U)} = \text{Multihead} (E^{(U)'}, E^{(U)'}, E^{(U)'}). \quad (11)$$

3.2.5. Hybrid. After the event is updated in conjunction with the GNN, the output of the pretrained S_2 layer is mixed with the event update result, and the attention mechanism is also used, and the specific method is as in Formulas (12) and (13):

$$H^{(S_2)^*} = \text{Multihead} (H^{(S_2)}, E^{(U)}, E^{(U)}), \quad (12)$$

$$H^{(S_2)^*} = \text{Layer norm} (H^{(S_2)^*} + H^{(S_2)}). \quad (13)$$

Put the result obtained by Formula (13) into the remaining k -layer conversion, and then the result as in Formula (14) can be obtained:

$$H = \text{Transformer}_{(s_2 \rightarrow k)} (H^{(S_2)^*}). \quad (14)$$

3.2.6. Prediction. Events after pretraining and GNN processing will be modeled as a binary description, and its specific formula is shown in Equation (15):

$$\text{Prediction} = \text{Argmax} (H_{[\text{CLS}]} W_p + b_p). \quad (15)$$

The Argmax function is used to obtain the argument that maximizes the value within the parentheses.

Among them, prediction is divided into 1 or 0, 1 means that the alternative event represents the follow-up of the context, and 0 means that the alternative event does not represent the follow-up of the context [28]. $H_{[\text{CLS}]}$ represents a vector of characters, $W_p \in \mathbb{R}^{d \times 2}$ is a trainable parameter, and b_p is an offset term.

3.3. Simulation Verification

3.3.1. Setting Up a Simulation Environment. In this paper, the MCNC event set is used for experiments. The event data are provided by Ant Financial Services and has the same features as described in Section 3.3 [29]. The specific quantity and number of instances in the dataset are shown in Table 1.

TABLE 1: Collection of MCNC events.

Metric	Training set	Test set	Development set
Number of texts	829,981	105,031	102,969
Number of event nodes	139,586	9896	9897

3.3.2. Simulation Settings. The simulation will build an event map on the training set of the event set and then verify the performance of the model on the event set. The loss generated during the simulation process is mainly composed of fitting loss and prediction loss. Since the edge loss is used to measure the prediction loss, the KL divergence configuration matrix is used to fit the loss. The specific conditions are shown in Formulas (16)–(18) composition:

$$L_{\text{pred}} = \max(0, \text{margin} - p_{\theta}(Y_i|X, z) + p_{\theta}(Y_j|X, z)), \quad (16)$$

$$L_a = KL(q(z|X, Y) \| p_{\theta}(z|X)), \quad (17)$$

$$L(\theta) = L_{\text{pred}} + \lambda L_a. \quad (18)$$

Using a pretrained model with 12 layers of conversion layer, which includes 768 dimensions, 12 head attention mechanism uses the 8th layer output of the pretrained model as the input of the pooling step and uses the 10th layer of the pretrained model as the output of the mixing step. The margin loss is set to 0.04.

3.3.3. Baseline Method. Word2Vec [30] trained word vector representations on a large-scale dataset, utilizing vectors of verbs and other contextual elements to compute similarity with elements in potential events, making predictions based on these similarity values.

Scaled GNN (SGNN) [31] established a narrative event evolution graph, referred to as the causal graph, illustrating connections between events and proposed a SGNN to forecast subsequent events.

BERT [32] performed fine-tuning on a pretrained BERT-based model using the MCNC dataset.

Graph transformer [33] extracted structured information from a knowledge graph and introduced an additional graph encoder at the top of their transformer-based model to assist in text generation guided by the retrieved structured information.

3.3.4. Simulation Results. Tested in all methods of the same type and other comparison methods of the same type includes random, Word2Vec, SGNN, BERT, and graph transformer. The obtained event prediction accuracy is shown in Table 2.

Through the accuracy results in Table 2, it can be found that:

1. Combining graph structure information can better complete event prediction.
2. Combining the pretrained model has a positive effect on event understanding and prediction.

TABLE 2: Event prediction accuracy rate.

Method	Prediction accuracy (%)
Number of random	21.01
Word2Vec	41.93
SGNN	52.48
BERT	56.02
Graph transformer	56.09
BERT + GNN	58.83

TABLE 3: Prediction accuracy corresponding to the loss of weight change.

λ	Accuracy (%)
0.010	59.17
0.015	58.43
0.020	59.55
0.025	58.78
0.030	58.89
0.035	58.41
0.040	58.68
0.045	58.02
0.050	58.02

TABLE 4: Relationship between different semantic complexity and prediction accuracy.

S_2	S_1	Accuracy (%)
9	5	57.84
10	5	58.20
10	5	58.30
11	6	58.19
10	6	58.61
11	7	58.66
10	8	58.02
11	7	58.21

- Combining the pretraining model with the graph structure information enhancement model can enhance the predictive performance of the model.

3.3.5. Ablation Experiment. In order to verify the feasibility and effectiveness of the method, the prediction loss and connection matrix loss weights are adjusted and tested, and it is found that the best event prediction accuracy can be achieved when the loss weight λ is 0.02. The details are shown in Table 3.

In addition, because different layers of the pretraining model will produce different information, using complex semantics and simple grammar to update the graph can significantly affect the event prediction results, so different semantic complexities S_1 and S_2 are used to complete the experiment. The experimental results are shown in Table 4. It is found that the best event prediction accuracy rate is obtained when the value is 10 and the value is 8.

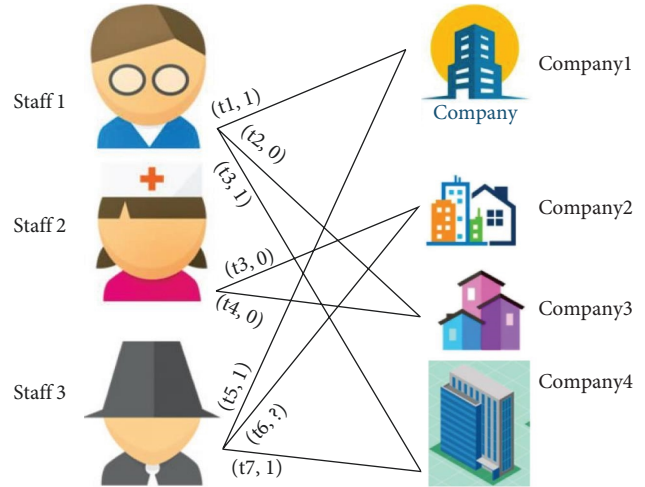


FIGURE 5: The relationship between employees and enterprises.

4. Experiment and Result Analysis

In this section, we will construct a scenario for employee career choice to validate the effectiveness of the proposed method. Firstly, we analyze the problem of employee career choice and then use a dynamic bipartite graph to solve the problem using the learning method. Finally, we quantitatively analyze the results obtained from the experiments.

4.1. Problem Analysis. Employee occupational choices are closely linked to the vitality and development of a company. Employee departures can negatively impact organizational growth, making it crucial to study their career decisions [34]. Currently, the prediction of employee career choice events largely depends on personal information, lacking analysis of social network footprint data. Thus, constructing a social network interaction information extraction model for employees holds significant practical value [35]. An employee's work history within an organization can be represented by a dynamic graph. Figure 5 illustrates the graph detailing the relationships between multiple companies and three employees. The links connecting enterprises and employees feature entry time-stamps and job status labels, where 0 and 1 are used as bipartite graph classification states to predict employees' occupational choices. For example, in Figure 5, employee u1 joined company1 at time t1 and later transitioned to company3 at time t2. The label at t1 is 1, indicating resignation, while the label at t2 is 0, indicating active employment [36].

After transforming the problem of employee career choice event prediction into a bipartite graph classification problem through the above methods, it is necessary to extract the characteristics of employee work records and enterprise nodes to classify. Specifically, it can be designed as a bipartite graph network $G = (X, Y, \mathcal{E}, \mathcal{T})$, where X and Y represent $\mathcal{E}$ which represents the set of relationship lines and $\mathcal{T}$ represents the entry time stamp on the line. For each line (staff, company, and t) linking staff and company with time stamp t , the employee's occupation choice prediction is defined as combining historical characteristics and network characteristics to

predict the turnover tendency of employees within a certain period of time after they join the job.

4.2. Dynamic Bipartite Graph Representation Learning Method. The script event prediction pertains to the homogeneous structure graph. To address the information extraction from the more intricate heterogeneous structure graph [37], this study employs the random walk method for simulating event node interactions and utilizes the dynamic bipartite graph representation learning method to scrutinize employee career choices.

Given the temporal sequence relationship among event nodes, which are indicative of employee resignations and new entries in their career decisions, this research amalgamates time-based random walk sampling to procure event nodes chronologically arranged. Subsequently, the skip-gram neural network model is employed to map nodes to low-dimensional vectors for representation purposes.

When employees resign from small-to-medium-sized private enterprises and transition to governmental or public institutions, historical employee resignations gravitate toward seeking more stable job opportunities with a relatively low likelihood of further resignations. Conversely, if employees resign to join larger Internet companies, they may be driven by higher remuneration prospects, thereby increasing the probability of future resignations.

In order to obtain the low-dimensional vector representation of the employee's career choice event, the skip-gram model is used to train the sampled sequence, and the time sequence obtained through the time walk is defined as $(v_1, v_2, v_3, v_4, v_5, v_6)$. The skip-gram model sequentially predicts the context of each node of the sequence; for example, the model optimization formula with an interval of 2 is $P(v_2|v_4) \cdot P(v_3|v_4) \cdot P(v_5|v_4) \cdot P(v_6|v_4)$.

When the context c of the specific sequence S and the central node w are known, after optimizing the conditional probability parameter $p(c|w; \theta)$, the maximum node sequence probability is obtained as

$$\arg \max_{\theta} \prod_{w \in S} \prod_{c \in \text{context}(w)} p(c|w; \theta),$$

it can also be abbreviated as Equation (19).

$$\arg \max_{\theta} \prod_{(w,c) \in D} p(c|w; \theta). \quad (19)$$

Among them, D represents the combination of context and intermediate points, and is described as Formula (20) in an accumulative way:

$$\arg \max_{\theta} \sum_{(w,c) \in D} \log p(c|w; \theta). \quad (20)$$

The definition of conditional probability becomes the key, and the main conditions include two:

$$0 \leq (c|w; \theta) \leq 1, \quad (21)$$

$$\sum_{c \in \text{context}(w)} \log p(c|w; \theta) = 1. \quad (22)$$

Using the softmax formula to normalize, Formula (23) can be obtained:

$$p(c|w; \theta) = \frac{e^{u_c \cdot v_w}}{\sum_{c' \in U} e^{u_{c'} \cdot v_w}}. \quad (23)$$

Among them, $v_w, u_c \in \mathbb{R}^d$ represents the vector representation of w and c , respectively, and U represents the available combination of context nodes. For similar nodes, the objective function can be described in Formula (24):

$$L(\theta) = \arg \max_{\theta} \sum_{(w,c) \in D} \left[\log \sigma(u_c \cdot v_w) + \sum_{c' \in N(w)} \log \sigma(-u_{c'} \cdot v_w) \right]. \quad (24)$$

The objective function uses parameter optimization to shorten the distance between similar event nodes and at the same time push away the distance between dissimilar nodes, so that the objective function reaches the optimal value.

4.3. Prediction Experiment of Employees' Career Choice Events. In predicting enterprise employees' career choice events, common machine learning evaluation metrics such as F1, R, P, and Acc are utilized for assessment. The parameters for the dynamic bipartite graph representation learning experiment are configured as follows: the number of iterations r is set to 80, the interval size w is 5, the maximum walk length l is stipulated as 15, and the output node feature dimension d is specified as 128.

A bipartite graph is employed to depict the learning model training process, generating a 128-dimensional feature vector representation for each event node. Subsequently, by reducing the feature dimensions to 1–9, the learned network features are validated to assess the impact on test results across various dimensionalities. To mitigate variance effects, the experiment is conducted 50 times, with the average value of each index being recorded. Test results demonstrate the optimization effects of diverse common indicators across different dimensions as illustrated in Figures 6–9.

5. Conclusions

Forecasting the developmental trends of events has been a longstanding focal point within academic circles. This paper strategically designs prediction models targeting two event categories: homogeneous structure events and heterogeneous structure events, followed by an empirical examination of their practical applications. The study amalgamates pretraining models with GNNs to construct a script event prediction model and conducts comparative analyses with similar models to validate the model's superiority in terms of accuracy. Furthermore, a dynamic bipartite graph representation learning technique, utilizing the random walk rule, is devised and applied to the realm of employee career choice prediction.

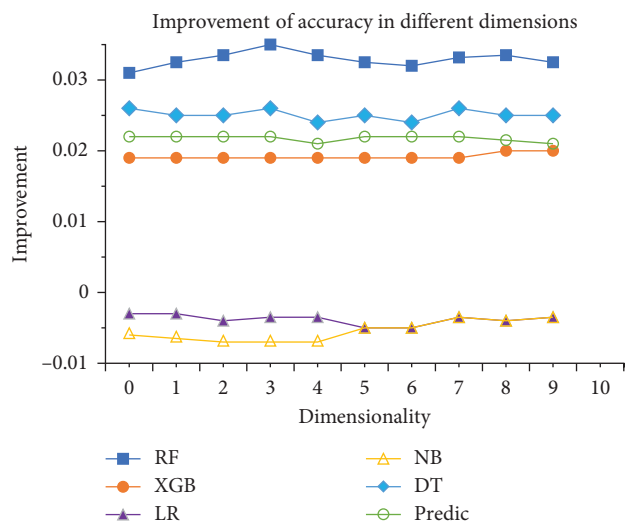


FIGURE 6: The optimization effect of accuracy in different dimensions.

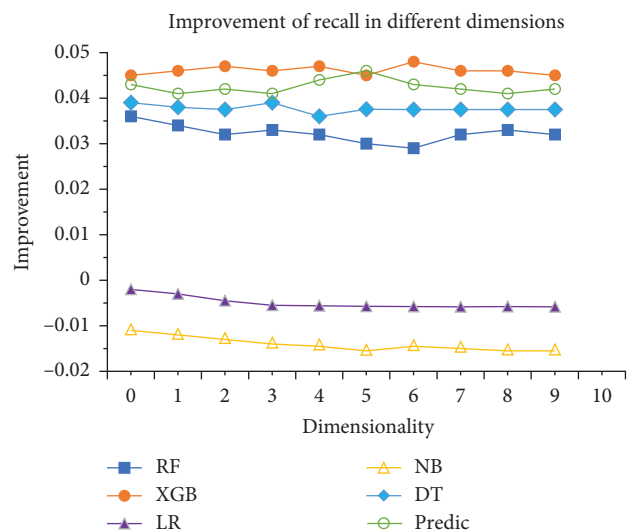


FIGURE 8: Optimization effect of recall rate under different dimensions.

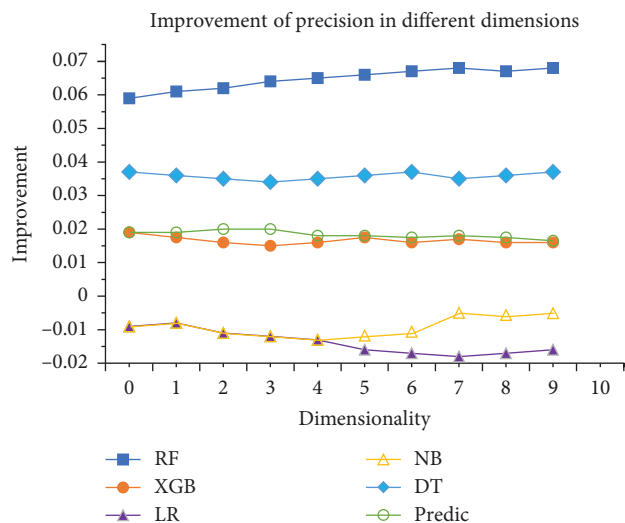


FIGURE 7: The optimization effect of precision rate in different dimensions.

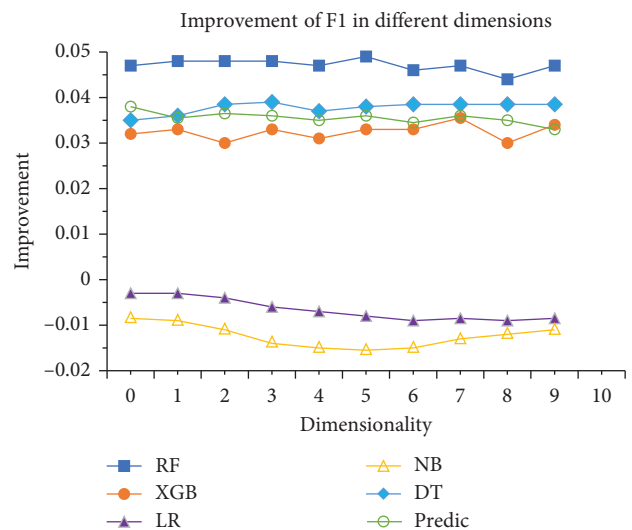


FIGURE 9: Optimization effect of F1 value in different dimensions.

Experimental outcomes demonstrate a significant enhancement in the predictive performance of employee career choices through this innovative approach.

While the model presented in this paper exhibits commendable performance in event prediction tests, there remains a necessity to fortify its robustness. Additionally, the model's applicability is relatively confined to a specific dataset type, thus indicating potential expansion requirements for its future scope of application. Through multisource data fusion and intelligent algorithms, event prediction technology can be widely used in more than 10 fields such as financial risk control, disease early warning, and traffic scheduling and is the core technical support for intelligent decision-making and risk prevention and

control. In the future, multimodal data (text, graph structure, and spatiotemporal series) will be combined to improve the prediction accuracy.

Data Availability Statement

The data that support the findings of this study are available upon request from the corresponding author. The data are not publicly available due to privacy or ethical restrictions.

Conflicts of Interest

The authors declare no conflicts of interest.

Funding

The authors received no specific funding for this work.

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